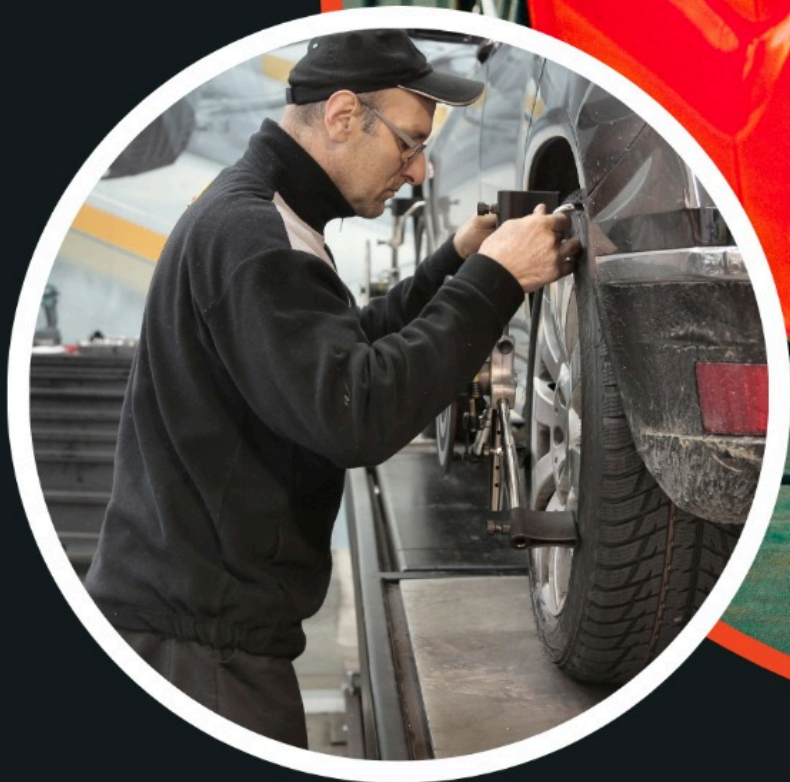




BACKLOG

SMART CAR SERVICE
BOOKING SYSTEM



GROUP 24

Backlog Documentation

Smart Car Service Booking System

GROUP 24

GROUP MEMBERS

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2. Objective

To develop an AI-enhanced car service booking system using Agile Scrum practices to personalize service recommendations based on car history, mileage, and driving patterns.

3. Introduction

The Smart Car Service Booking System aims to provide users with a seamless and intelligent way to manage their vehicle maintenance. By leveraging AI, the system suggests appropriate service packages tailored to each user's needs. The platform enables easy booking, real-time service updates, and personalized notifications to ensure optimal car maintenance. Built using Agile Scrum practices, this project will demonstrate iterative development, continuous feedback, and team collaboration to create a user-centric solution.

4. User Stories & Prioritization

Our backlog -  Smart car service - Product Backlog - Group 24

Each user story is categorized based on priority:

- **High Priority (H):** Critical for core functionality.
- **Medium Priority (M):** Important but not essential in the first development phase.
- **Low Priority (L):** Additional features that can be implemented later.

A. User Registration & Car Profile Management

User Story	Priority
As a user, I want to register and add my car details so that I can receive personalized service recommendations.	H

As a user, I want to edit and update my car profile to reflect changes in ownership or vehicle condition.	M
As a user, I want to store multiple car profiles under one account to manage all my vehicles easily.	L

B. AI-Driven Service Recommendations

User Story	Priority
As a user, I want AI-driven suggestions for car services based on mileage and past maintenance records so that I can keep my car in good condition.	H
As a user, I want AI-generated maintenance alerts to notify me when a service is due, preventing unexpected breakdowns.	H
As a user, I want recommendations to be refined based on my driving habits so that I receive relevant service suggestions.	M

C. Service Booking

User Story	Priority
As a user, I want to view available service slots so that I can choose a convenient time for my appointment.	H
As a user, I want to select a preferred service center from nearby locations so that I can minimize travel time.	H
As a user, I want to receive a confirmation message after booking my service to ensure my appointment is secured.	H

D. Notifications & Updates

User Story	Priority
As a user, I want to receive reminders for upcoming services so that I don't forget important maintenance tasks.	H

As a user, I want real-time updates on my car service progress so that I stay informed throughout the process.	M
As a user, I want to get notified when my car is ready for pickup so that I can plan accordingly.	M

E. Admin Dashboard

User Story	Priority
As a service provider, I want to manage and view service bookings so that I can optimize scheduling.	H
As a service provider, I want to update service status in real-time so that customers remain informed.	L
As a service provider, I want access to customer history and preferences to provide better service recommendations.	L

F. Payment & Billing

User Story	Priority
As a user, I want to pay for my service through multiple payment options.	H
As a user, I want to view my past service invoices.	M
As a user, I want a refund policy for canceled services.	L

5. Tasks & Technical Requirements

Each user story is broken down into specific development tasks.

A. Frontend Tasks

- Design login & registration UI
- Implement form validation
- Create a car profile management interface
- Develop a booking page with a calendar selection
- Implement real-time notifications & status updates

B. Backend Tasks

- Set up user authentication (JWT-based)
- Develop REST API for car data management
- Implement an AI-based recommendation system
- Develop service booking & availability logic
- Set up payment gateway integration

C. Testing & QA

- Unit Testing – Login, Booking, AI Suggestions
- API Testing – AI Recommendations, Payment Processing
- Integration Testing – End-to-end flow testing
- Load Testing – High-traffic booking scenarios

6. Backlog Management

User stories were created based on common customer pain points and business requirements. The backlog was continuously refined using Agile Scrum techniques, ensuring that high-priority tasks were completed first. Key steps included:

- Identifying core functionalities essential for a car service booking system.
- Prioritizing features based on user needs and business impact.
- Refining the backlog after sprint reviews and customer feedback.

Our tracker -  Smart car service - Product Backlog - Group 24

Smart car service - Product Backlog - Group 24						
File Edit View Insert Format Data Tools Extensions Help						
Q Menus 100% \$ % .00 123 Raleway 11 B I A						
A1	Epic					
1	Epic	Priority	Assigned To	Status	Start Date	End Date
2	User Registration & Car Profile Management					
3	As a user, I want to register and add my car details so that I can receive personalized service recommendations.	H	Chathumini	Not Complet	14th Feb	16th Feb
4	As a user, I want to edit and update my car profile to reflect changes in ownership or vehicle condition.	M	Chathumini	Not Complet	14th Feb	16th Feb
5	As a user, I want to store multiple car profiles under one account to manage all my vehicles easily.	L	Chathumini	Not Complet	14th Feb	20th Feb
6						
7	AI-Driven Service Recommendations					
8	As a user, I want AI-driven suggestions for car services based on mileage and past maintenance records so that I can keep my car in good condition.	H	Farshad	Not Complet	15th Feb	16th Feb
9	As a user, I want AI-generated maintenance alerts to notify me when a service is due, preventing unexpected breakdowns.	H	Farshad	Not Complet	15th Feb	16th Feb
10	As a user, I want recommendations to be refined based on my driving habits so that I receive relevant service suggestions.	M	Farshad	Not Complet	15th Feb	16th Feb
11						
12	Service Booking					
13	As a user, I want to view available service slots so that I can choose a convenient time for my appointment.	H	Kasundie	Not Complet	15th Feb	22nd Feb
14	As a user, I want to select a preferred service center from nearby locations so that I can minimize travel time.	H	Kasundie	Not Complet	15th Feb	22nd Feb
15	As a user, I want to receive a confirmation message after booking my service to ensure my appointment is secured.	H	Kasundie	Not Complet	15th Feb	19th Feb
16						
17	Notifications & Updates					
18	As a user, I want to receive reminders for upcoming services so that I don't forget important maintenance tasks.	H	Bhagya	Not Complet	15th Feb	16th Feb
19	As a user, I want real-time updates on my car service progress so that I stay informed throughout the process.	M	Bhagya	Not Complet	15th Feb	17th Feb
20	As a user, I want to get notified when my car is ready for pickup so that I can plan accordingly.	M	Bhagya	Not Complet	15th Feb	16th Feb
21						
22	Admin Dashboard					
23	As a service provider, I want to manage and view service bookings so that I can optimize scheduling.	H	Sadewni	Not Complet	15th Feb	17th Feb
24	As a service provider, I want to update service status in real-time so that customers remain informed.	L	Sadewni	Not Complet	15th Feb	17th Feb
25	As a service provider, I want access to customer history and preferences to provide better service recommendations.	L	Sadewni	Not Complet	15th Feb	18th Feb
26						
27						

7. Sprint Planning

The project was divided into two sprints:

- Sprint 1 (Foundational Development)
 - User registration & car profile management
 - AI-driven service recommendations (basic implementation)
 - Service booking system (appointment scheduling)
- Sprint 2 (Enhancements & AI Integration)
 - Advanced AI-driven recommendations based on driving habits
 - Notifications & updates for service status
 - UI improvements and optional admin dashboard

Sprint priorities were set based on dependencies, ensuring that user registration and booking functionalities were available before AI-driven recommendations.

8. Scrum Events

- Daily Stand-ups: Short meetings to discuss progress, blockers, and task assignments.
- Sprint Reviews: Demonstrating completed work to stakeholders for feedback.
- Sprint Retrospectives: Evaluating team performance and identifying areas for improvement.

9. AI Implementation

- AI models analyze car history, mileage, and driving habits to generate personalized service recommendations.
- Predictive maintenance alerts help users schedule services before issues arise.
 - AI continuously learns from user behavior to refine future recommendations.